ORBITAL MECHANICS OF PLANETS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Orbital Mechanics of Planets

Question 1

10.0 points

Examine the diagram below depicting a planet's elliptical orbit around the sun. Using Kepler's second law, describe how the area swept out by a line from the planet to the sun varies over time. Explain how the speed of the planet changes as it moves along different segments of its orbit and why.

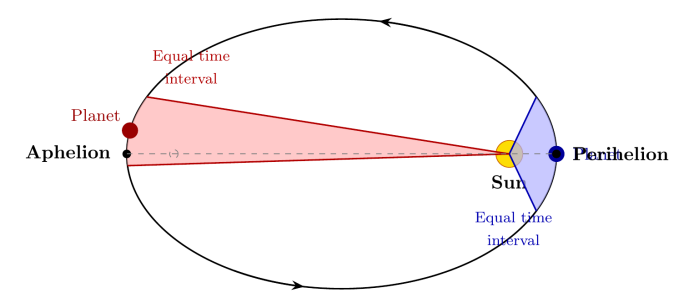

Fig. 1. An elliptical orbit of a planet around the sun. The sun is located at one focus of the ellipse. The diagram should include several segments of the orbit, each representing equal time intervals, with arrows indicating the direction of the planet's motion. Label the sun, the planet, and the segments of the orbit. Highlight the areas swept out by a line from the planet to the sun over these intervals to illustrate Kepler's second law.

Question 2 10.0 points

Refer to the diagram below, which illustrates a spacecraft executing a Hohmann transfer orbit between two circular orbits around Earth. Calculate the velocity changes (Δv) required at each segment of the transfer. Provide a detailed step-by-step explanation of the calculation.

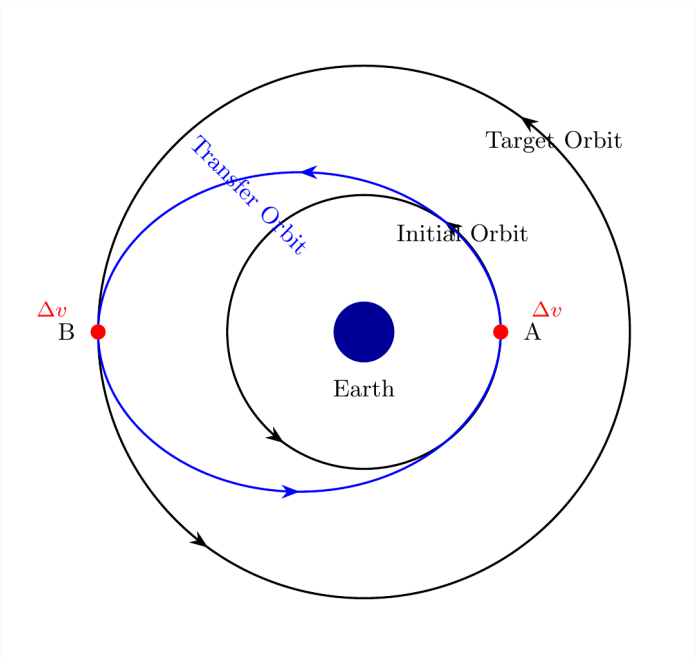

Fig. 2. MANDATORY LABELS — every item below MUST appear as visible text in the diagram: • Δv The diagram will FAIL automated review if any of these are missing. A Hohmann transfer orbit diagram showing two concentric circular orbits around Earth. The inner orbit is the initial orbit, and the outer orbit is the target orbit. An elliptical transfer orbit connects the two circular orbits, sharing one focus with the Earth. Mark the points of velocity change (Δv) at the periapsis and apoapsis of the elliptical orbit. Label the inner orbit as 'Initial Orbit', the outer orbit as 'Target Orbit', and the elliptical path as 'Transfer Orbit'. Include Earth at the common focus of the orbits.

Question 3 10.0 points

Refer to the diagram below illustrating the trajectory of a space probe executing a gravitational slingshot maneuver around a planet. Describe how this maneuver alters the probe's trajectory and velocity, and discuss the conservation laws that apply in this context.

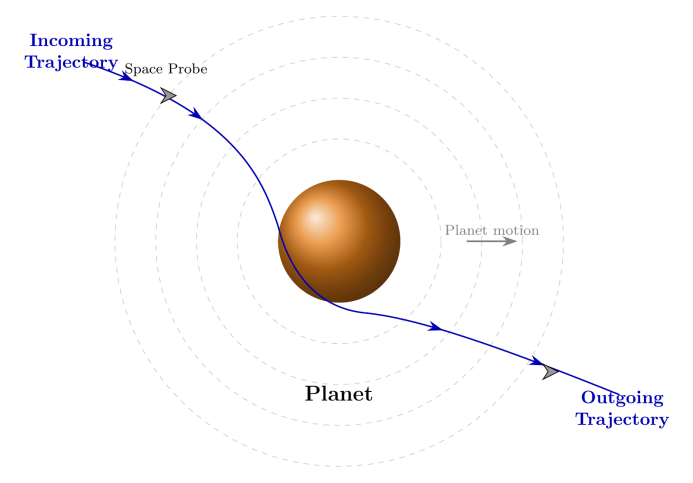

Fig. 3. A diagram illustrating the trajectory of a space probe executing a gravitational slingshot maneuver around a planet. The diagram should show the planet at the center with its gravitational field indicated by concentric circles. The space probe's trajectory should be depicted as a curved path approaching the planet, bending around it, and then exiting. Vector arrows should indicate the probe's velocity before and after the maneuver, showing an increase in velocity. Labels should include 'Incoming Trajectory', 'Outgoing Trajectory', 'Planet', and 'Velocity Change'.

Question 4 10.0 points

Examine the diagram below illustrating the five Lagrange points (L1-L5) in the Earth-Sun system. Discuss the stability of each point and explain how satellites can utilize these positions.

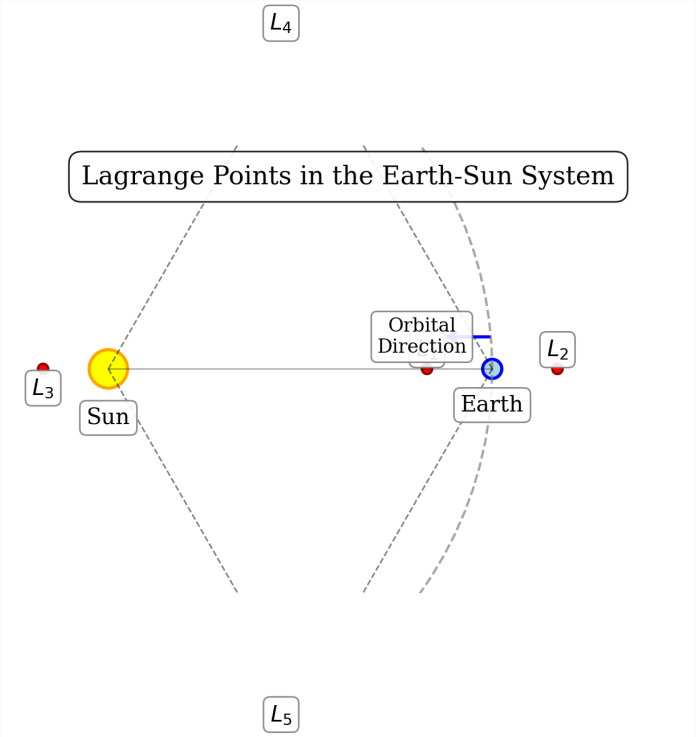

Fig. 4. Diagram illustrating the five Lagrange points (L1-L5) in the Earth-Sun system. The Sun is on the left, Earth is on the right, and the Lagrange points are labeled L1, L2, L3, L4, and L5. L1 is between the Earth and Sun, L2 is beyond Earth on the same line, L3 is on the opposite side of the Sun, and L4 and L5 form equilateral triangles with the Earth and Sun. Include arrows to indicate gravitational forces and show the relative positions of these points in space.

Question 5 10.0 points

Using the diagram below that details Mercury's orbit, explain the phenomenon of orbital precession and its causes, including contributions from both Newtonian mechanics and general relativity, with equations <eq_q_1> and <eq_q_2> resolved.

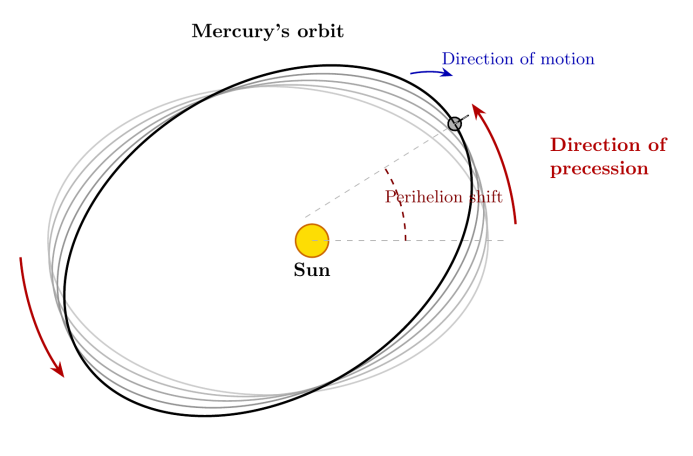

Fig. 5. Diagram showing Mercury's elliptical orbit around the Sun with a focus on orbital precession. The orbit should be depicted as an ellipse with the Sun at one focus. Indicate the direction of Mercury's motion and the gradual shift (precession) of the orbit over time. Include labels for key elements such as the Sun, Mercury's orbit, and the direction of precession. Use arrows to show the precession direction and annotate with relevant angles or distances if needed to illustrate the concept of precession.